

Supplementary Material: Auditing Annotation Support and Evaluation Sensitivity in Aerial Vehicle Detection

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TABLE I
NOTATION USED IN THE SUPPLEMENTARY ANALYSIS.

Symbol	Definition
$G_s, G_s(r)$	Valid mapped boxes; boxes retained by rule r
$S_s(r)$	Annotation-support shift $1 - G_s(r) / G_s $
$m_k(r), B_k$	Candidate metric; within-policy metric band
$\Delta_j(r)$	Winner-competitor margin under rule r
Γ_j	Maximum change of Δ_j from reference r_0
C_s	Source-wise common image-coverage denominator
T, P, N	Global TP, FP, and FN counts for micro- F_1
D	$2T + P + N$, the micro- F_1 denominator

ι is IoU. AP integrates over score order and hence has no fixed confidence coordinate. All multi-candidate comparisons are computed on C_s , defined before class filtering from raw-cache image names. A covered image with no retained vehicle prediction remains an evaluated empty prediction.

S1. NOTATION AND AUDIT QUANTITIES

This supplement specifies the formal objects and the data-free records underlying the manuscript. All reported quantities are restricted to the declared artifacts, mapping, common-coverage denominator, frozen candidate pool, and registered rule set. They are not estimates of annotation-error rates, a general detector leaderboard, or physical traffic density.

Let G_s be the valid mapped ground-truth boxes for source s , and let $G_s(r) \subseteq G_s$ be the boxes retained by a support rule r . The annotation-support shift is

$$S_s(r) = 1 - \frac{|G_s(r)|}{|G_s|}.$$

For candidate k and metric m , a policy band at fixed scale, confidence, and IoU is $B_k = \max_{\pi \in \Pi_s} m_k(\pi) - \min_{\pi \in \Pi_s} m_k(\pi)$. The authoritative primary policy set is $\Pi = \{I, O, N\}$: include-all (I), exclude/source-ignore-off (O), and exclude/source-ignore-on (N). The ancillary “exclude-and-ignore-removed” control is intentionally outside this set and therefore does not enter bands, headline record counts, or Fig. 2 of the main paper.

For a reference rule r_0 , winner k^* , and competitor j , define $\Delta_j(r) = m_{k^*}(r) - m_j(r)$ and

$$\Gamma_j = \max_{r \in \mathcal{R}} |\Delta_j(r) - \Delta_j(r_0)|.$$

Γ_j is the differential policy radius. It removes common-mode score movement and is the relevant deterministic diagnostic for a pairwise rank claim. We additionally report $\min_{r \in \mathcal{R}} \Delta_j(r)$, the direct observed rank margin, and a paired-bootstrap winner probability for the declared resampling unit.

The main paper’s rule tuple is $r = (d, \tau, \pi, c, \iota)$, where d selects absolute or normalized support, τ is the support threshold, π is a small-object policy, c is confidence, and

TABLE II
COMPUTED PAIRWISE DIAGNOSTICS FROM EXISTING HEADLINE RECORDS
(SIX NOMINAL POLICIES PER ROW).

Source	Metric	Δ_0	Γ	$\min \Delta$	Result
VisDrone	AP ₅₀	.3544	.0206	.3544	stable
Local UAVDT	AP ₅₀	.0340	.0326	.0014	deterministic only
VisDrone	F_1	.3354	.0026	.3354	stable
Local UAVDT	F_1	.1911	.00007	.1910	deterministic only

S2. FORMAL RESULTS AND PROOFS

Lemma S1 (resize invariance). Under isotropic resizing by factor $\alpha > 0$, (w, h, W, H) maps to $(\alpha w, \alpha h, \alpha W, \alpha H)$. Thus $s'_a = \max(\alpha w, \alpha h) = \alpha s_a$, while

$$s'_n = \max\left(\frac{\alpha w}{\alpha W}, \frac{\alpha h}{\alpha H}\right) = s_n.$$

Consequently, a fixed-pixel threshold depends on image scale, whereas a normalized-side threshold is invariant to isotropic image resizing. This is an invariance statement about the audit quantity, not a claim that normalized filtering removes every source difference.

Proposition S1 (policy-robust candidate-pool winner). Suppose k^* is the reference winner and $\Delta_j(r_0) > 0$ for every competitor j . If $\Delta_j(r_0) > \Gamma_j$ for all j , then k^* is the winner for every $r \in \mathcal{R}$.

Proof. By the definition of Γ_j ,

$$\Delta_j(r) \geq \Delta_j(r_0) - |\Delta_j(r) - \Delta_j(r_0)| \geq \Delta_j(r_0) - \Gamma_j > 0.$$

Thus $m_{k^*}(r) > m_j(r)$ for every competitor and every registered rule. $\square$

Corollary S1. Since

$$\Gamma_j \leq B_{k^*} + B_j,$$

the sufficient condition $\Delta_j(r_0) > B_{k^*} + B_j$ guarantees the same conclusion.

Proof. Write $x_k(r) = m_k(r) - m_k(r_0)$. Each $x_k(r)$ lies between the minimum and maximum movement of candidate k , whose range is B_k . Hence $|x_{k^*}(r) - x_j(r)| \leq B_{k^*} + B_j$, and maximization over r gives the inequality. $\square$

The corollary is deliberately conservative: it ignores correlation in rule-induced movements. For example, let $(m_A(r_0), m_B(r_0)) = (.70, .68)$ and, under another rule, $(.60, .58)$. Both individual bands equal .10, exceeding the reference margin .02, yet $\Delta_B(r) = .02$ throughout and $\Gamma_B = 0$. Conversely, anti-aligned movements can make small individual bands consequential for a close race. Differential radii therefore complement, rather than replace, the individual policy bands.

S3. COVERAGE IDENTIFIABILITY AND AN F_1 BOUND

Proposition S2 (non-identifiability under unequal coverage).

Let candidates A and B be evaluated on different image subsets C_A and C_B . Without assumptions on unobserved performance on $C_B \setminus C_A$ and $C_A \setminus C_B$, their ordering over the full evaluation population is not identifiable from observed subset metrics.

Proof. Keep all observed predictions and annotations fixed. On the unobserved portion, construct one completion where A is assigned sufficiently favorable outcomes and B sufficiently unfavorable outcomes; this yields an A -leading full-population metric while preserving every observed subset metric. Construct a second completion with the assignments reversed; it yields a B -leading metric and the same observed data. Since both orderings are compatible with the observations, no unique full-population ordering is identified. $\square$

This result motivates the source-wise common denominator used for VisDrone and local UAVDT. It also gives the formal boundary for AI-TOD: its 1,869 covered images have 16,900 vehicle boxes (9.0 per image; median side 12 px), while 935 noncovered images have 43,004 boxes (46.0 per image; median side 16 px). The observed coverage-conditioned sensitivity analysis cannot be extrapolated into a full-source ranking without missing-coverage assumptions.

Proposition S3 (count-perturbation bound for micro- F_1).

For $F(T, P, N) = 2T/(2T + P + N)$, let $D_{\min} > 0$ lower-bound $D = 2T + P + N$ along the segment between two nonnegative count vectors. Then

$$|\Delta F| \leq \frac{2|\Delta T| + |\Delta P| + |\Delta N|}{D_{\min}}.$$

Proof. The partial derivatives satisfy $|\partial F/\partial T| = 2(P + N)/D^2 \leq 2/D$, and $|\partial F/\partial P| = |\partial F/\partial N| = 2T/D^2 \leq 1/D$. Apply the integral form of the mean-value bound along the line segment and use $D \geq D_{\min}$. $\square$

The bound does not predict an observed band from support removal alone, because the signs and magnitudes of TP, FP, and FN changes remain detector- and policy-dependent. It does explain why extreme removal permits a large band: at 24 px, the pooled excluded fractions are 26.7%, 4.9%, and 90.3% for VisDrone, local UAVDT, and AI-TOD, respectively. AI-TOD's .330 F_1 band is therefore interpreted jointly with retained support and removed-object treatment, not as an isolated detector effect.

TABLE III
ARTIFACT-LEVEL PROVENANCE AND EVALUATION SCOPE.

Source	Eval. imgs	Covered	Vehicle boxes	Mapping	Scope
VisDrone	548	548	17,040	car/van/truck/bus	full audit
Local UAVDT	305	305	11,054	car/truck/bus	full audit
AI-TOD	2,804	1,869	59,904	vehicle	sensitivity only
DOTA-v2	5,108	–	53,349	small/large vehicle	structural only

S4. ARTIFACT PROVENANCE AND COVERAGE AUDIT

Table III records the source-specific boundaries that matter for interpretation. DOTA-v2 contributes only structural support: its quadrilaterals are converted to axis-aligned horizontal-box proxies for the scale audit, so its distances are representation-conditioned. The local UAVDT artifact is a COCO-style conversion and is conversion-conditioned. AI-TOD’s available crop reconstruction is xView-derived and has incomplete prediction coverage; it is not used for a rank claim.

The vehicle-positive image counts are 519/548 for VisDrone, 305/305 for local UAVDT, 1,443/2,804 for AI-TOD, and 2,581/5,108 for DOTA-v2. Median horizontal-box maximum sides are 41, 33, 14, and 18 px, respectively; median normalized sides are .0412, .0278, .0175, and .0352. These are descriptive support properties. They do not establish which source is “more correct,” nor do they assert a causal resolution effect.

Raw prediction coverage is determined by image-name presence before mapping predictions into the vehicle superclass. Consequently, a covered image whose predictions are entirely filtered out still contributes all valid mapped ground truth as false negatives, rather than vanishing from AP or F_1 . Source ignores are preserved for VisDrone. Its truncation and occlusion attributes are not filtered. For local UAVDT, the applied coordinates are the recorded *xywh* conversion. The provenance manifest in the repository supplies raw paths and hashes without redistributing licensed images or prediction caches.

The published package at <https://github.com/Marchematics/grsl-aerial-evaluation-audit> contains the ontology mapping, candidate manifest, coverage records, corrected headline outputs, matching controls, bootstrap records, figure-source summaries, scripts, and SHA-256 manifest. It is sufficient to inspect every claim-scoped derived record; acquiring the licensed source artifacts remains the responsibility of the reproducer.

TABLE IV
POLICY SEMANTICS AND ROLE IN REPORTED QUANTITIES.

Policy	Removed valid boxes	Use
Include-all	retained	primary grid
Exclude/source-ignore-off	absent; overlaps may be FP	primary grid
Exclude/source-ignore-on	absent; source ignores suppress overlaps	primary grid
Exclude-and-ignore-removed	absent; removed overlaps suppressed	ancillary diagnostic

S5. FULL RULE CONTRACT AND MATCHING CONTROLS

The full greedy surface comprises seven candidates, eight confidence values, four IoUs, eight absolute or six normalized support thresholds, and three primary policies: $7[8 \cdot 4 \cdot 8 \cdot 3 + 8 \cdot 4 \cdot 6 \cdot 3] = 9,408$ records. The absolute thresholds are $\{16, 20, 24, 28, 32, 40, 48, 64\}$ px, normalized thresholds are $\{.005, .0075, .010, .015, .020, .030\}$, confidence values are $\{.10, .15, .20, .25, .30, .35, .40, .50\}$, and IoUs are $\{.25, .30, .40, .50\}$. Headline rules use 24 px and .015, registered interior grid values selected before headline computation.

For fixed-operation F_1 , predictions are score ordered. Each prediction matches the highest-IoU eligible yet-unmatched valid ground-truth box above the declared IoU; unmatched predictions are FP and unmatched retained boxes are FN. The Hungarian control first removes invalid edges, maximizes cardinality of valid matches, and then maximizes total IoU among maximum-cardinality solutions. Dummy nodes permit unmatched detections and boxes. This threshold-first lexicographic construction prevents a high-IoU but sub-threshold assignment from displacing two valid matches.

AP uses one-category COCO bbox evaluation with complete score order, IoU .5 for AP_{50} , standard area ranges, and registered $\max Dets = 100$. The 100-detection cap is material: after source-ignore handling, it truncates predictions in 406–490 VisDrone images and 272–292 local-UAVDT images across primary policies. Focused 300 and 2000 controls preserve every one of the 10 nonduplicate multi-candidate AP winners; at 2000 no retained prediction is truncated (the largest retained count is 1,066), but AP_{50} pairwise gaps relative to 100 shift by $[-.041, .048]$.

The ancillary control is especially informative for AI-TOD: under it, F_1 is .130 at 24 px and .317 at .015, compared with .065 and .260 under exclusion without protection. Thus the observed policy variation combines extreme support removal and the treatment of predictions that overlap removed targets.

TABLE V
REPRODUCIBILITY AND CLAIM-AUDIT CONTRACT.

Item	Released or fixed record
Rule contract	configs/locked_rule_grid.yaml
Ontology/mapping	label_reliability/config/ontology.json
Coverage denominator	outputs/coverage/*.csv
AP/F1 headline records	outputs/ap_headline, coverage_corrected_grid
Matcher control	outputs/matching_headline/*.csv
Bootstrap outputs	outputs/statistics/bootstrap_headline/*.csv
Policy diagnostics	pairwise_policy_robustness.csv
Integrity check	SHA256SUMS; claim audit JSON

S6. BOOTSTRAP, REPRODUCIBILITY, AND CLAIM AUDIT

Pairwise uncertainty resamples the common evaluation unit jointly for the leading and runner-up candidates. VisDrone uses image units. Local UAVDT uses a filename-derived sequence proxy (substring before the first underscore): 93 units over 305 images, with min/median/max unit sizes of 1/1/104. Each analysis uses 1,000 paired replicates, seed 20260712, and percentile 2.5/97.5% intervals. Replicates recompute global micro- F_1 from resampled TP/FP/FN totals and recompute score-ordered one-category COCO AP_{50} at $\text{maxDets} = 100$.

For AP_{50} , all six VisDrone policy-specific intervals favor SAHI: their lower endpoints range from .339 to .362 and their upper endpoints from .366 to .387. Every one of the four nonduplicate local-UAVDT sequence-proxy intervals contains zero, with lower endpoints from $-.310$ to $-.296$, upper endpoints .071–.111, and winner probabilities .471–.604. For F_1 , all 12 VisDrone intervals are positive with winner probability 1.00, whereas all 12 local-UAVDT sequence-proxy intervals contain zero despite deterministic tiling wins (winner probabilities .691–.811). The latter unit is a sensitivity device, not evidence that filename prefixes recover official UAVDT sequence identities.

A minimal reproduction order is: verify hashes; inspect coverage and the frozen candidate manifest; evaluate AP and coverage-corrected absolute/normalized grids; run threshold-first Hungarian and removed-as-ignore controls; run paired bootstraps; regenerate the pairwise-policy diagnostic; and compile the manuscript. The data-free review bundle deliberately omits licensed raw imagery and prediction Parquet files while retaining their manifest paths and hashes. The automated manuscript claim audit verifies 43 numerical claims against released derived records. This makes the scope of each figure, table, and headline statement inspectable without extending the declared candidate pool or rerunning model training.